Group symbol: **3**

Team: **1**

Project title: **Resource Allocation App**

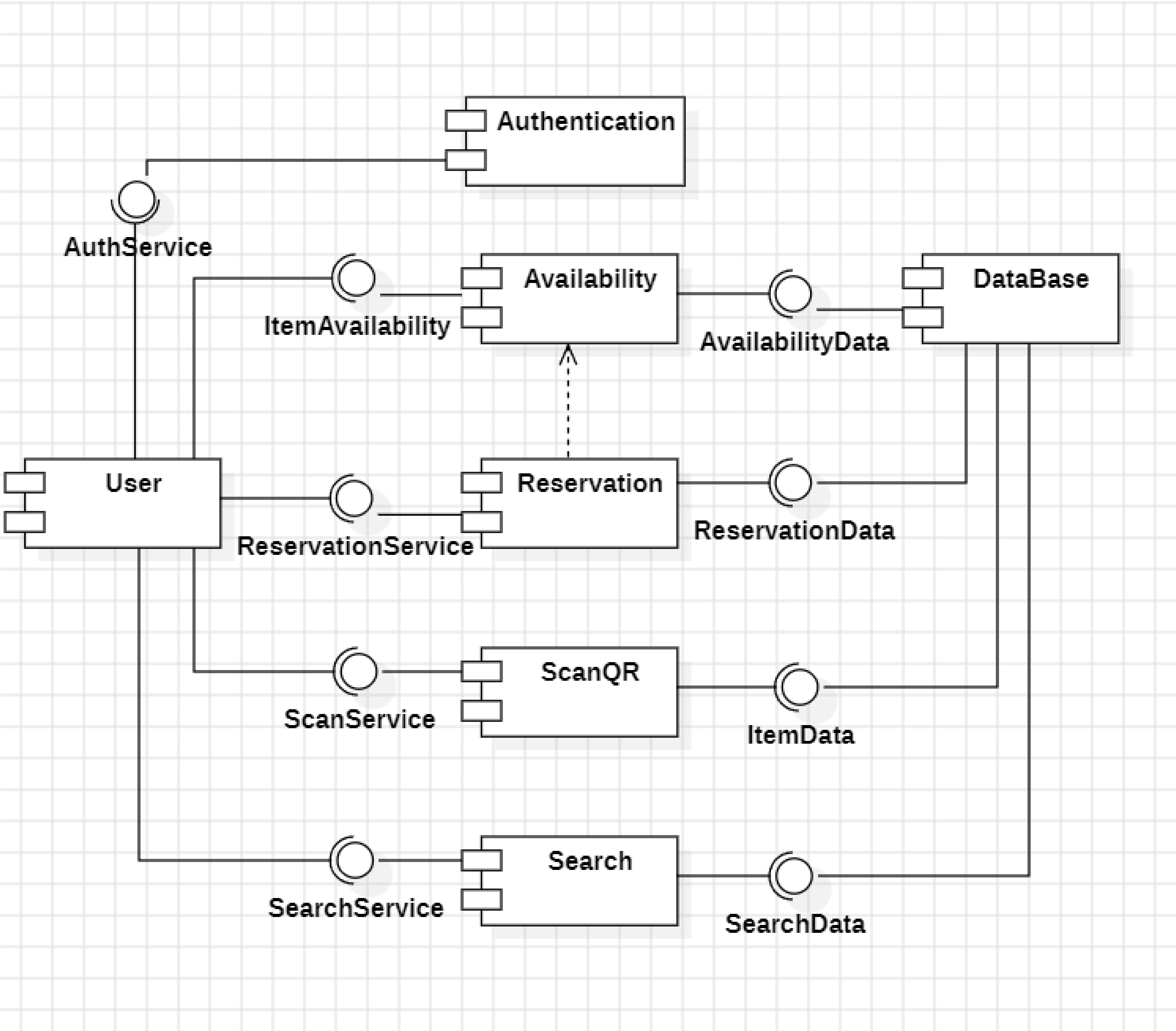
Team members (*filled by PM, Team Leader*):

No	Name	Surname	Student ID	Role
1	Carmelo Aidan	Romero	293936	*PM, Team Leader*
2	Álvaro	Puebla	293867	*Team member*
3	David	Sosa	293885	*Team member*
4	Diego	Ceballos	293943	*Team member*
5	Diego	García	293939	*Team member*
6	Óscar	López	293871	*Team member*

3. Design (S3)

3.1. Logical Software Architecture

The logical architecture aspect of the system should be present as a Component Diagram expressed in UML 2.5.

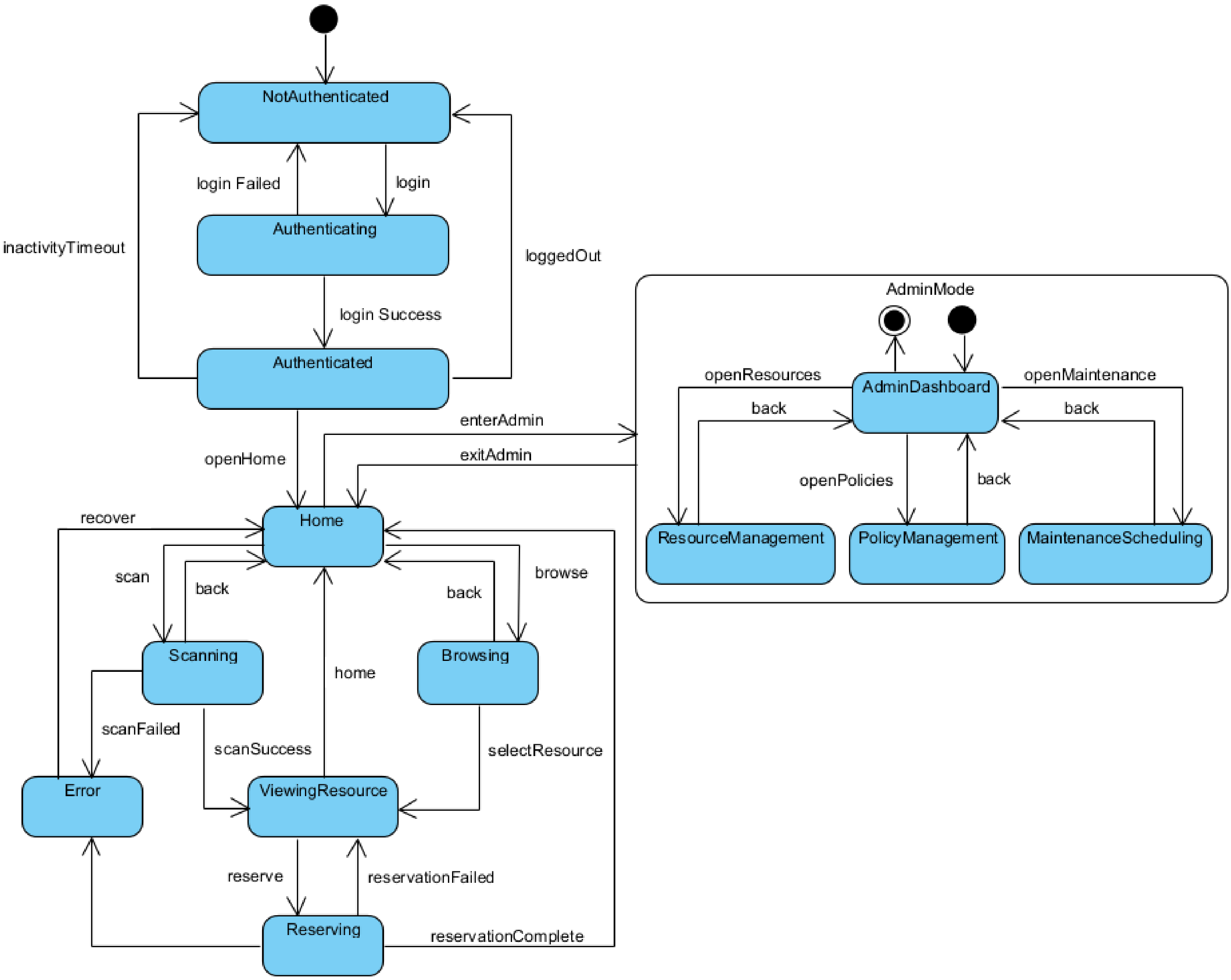


3.2. Business Logic Model

3.2.1. Behavioural Model

In this section present:

- *state diagram for whole the system and*



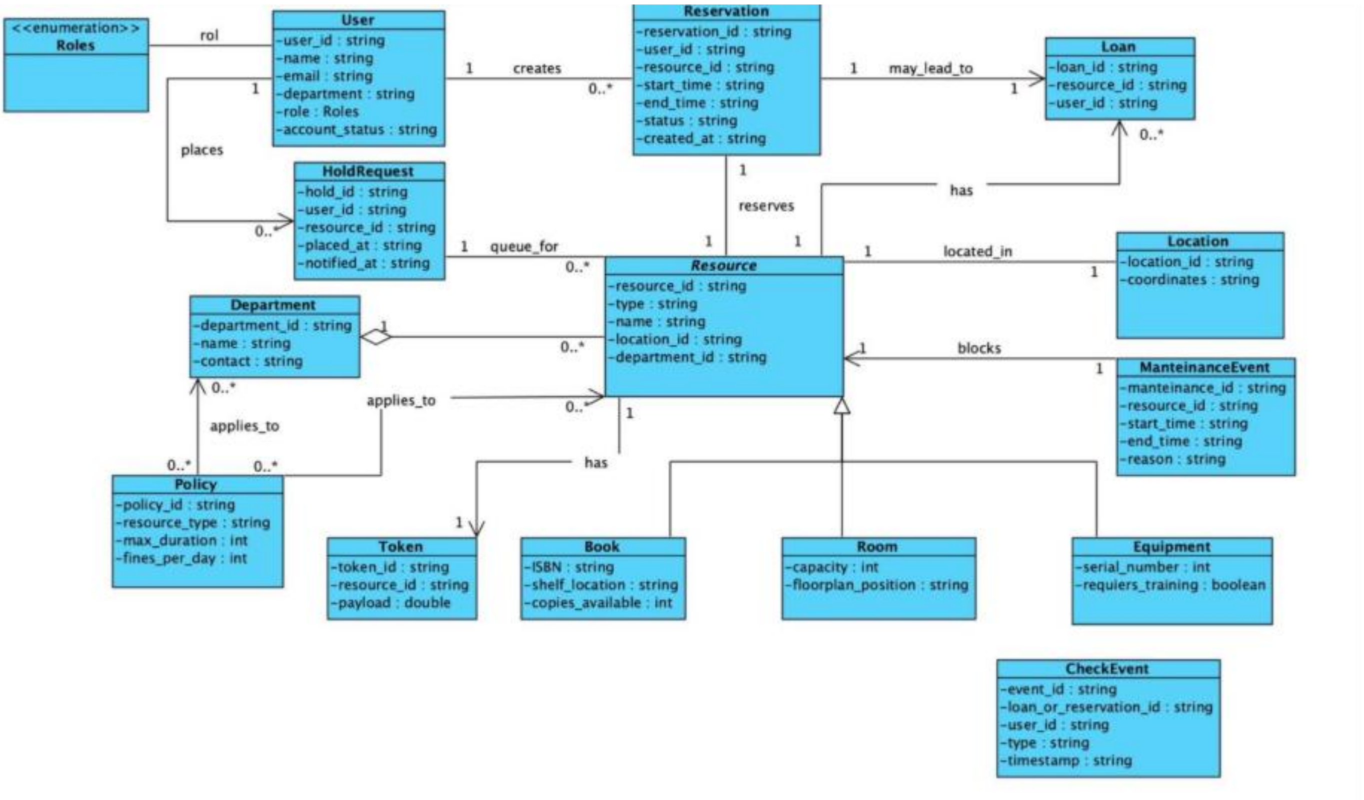
- *behaviour of 2 crucial use cases using the UML 2.5 diagrams, namely sequence diagram, and activity diagram.*

UC01	Making a reservation
Description	The client does a reservation for a book
Prerequisites	The client has logged in
Normal flow	
1	The client provides the information needed to search the book.
2	The system shows a list of books and their current availability.
Variant	
1.2	The client scans the QR code of the book with the app.
2.2	The system shows the book and its availability.
3	The client selects a book to reserve
4	The client makes the reservation.
5	The system confirms the reservation

UC02	Cancelling a reservation
Description	The client cancels the reservation and returns the book.
Prerequisites	The client has logged in and has a reservation.
Normal flow	
1	The client scans the QR code of the book with the app.
2	The system shows the reservation.
3	The client cancels the reservation and returns the book.
4	The system confirms the cancellation and updates the book’s availability.

3.2.2. Structural Model

Prepare the model as class diagram expressed in UML 2.5. Use natural language to describe the semantics of models' elements. Besides, use the object diagram to present an instance of the model at the time of system execution.



3.3. Database Model

3.3.1. Conceptual Model

The model contains the definition of entities and relationships among them in terms of the database (persistence). You should present it as a diagram and description of key elements in natural language.

This model represents a reservation system that manages different kinds of resources and the users who interact with them. It includes two specialization hierarchies and clearly defines how reservations are created, controlled, and stored.

1. Entities

User

Represents any person registered in the system.
Key attributes include an identifier, name, contact information, credentials, account creation date, and account status. This entity functions as the parent for two specialized roles: **Client** and **Admin**.

Client

A specialized type of user who can make reservations and loans.
Additional attributes define limits on active reservations and loans, the type of client, and an optional date until which the client may be blocked.

Admin

Another specialization of the User entity, representing staff members who oversee the reservation process.
Attributes include an employee code, administrative level, department information, and an indication of whether the account is active.

Resource

Represents any item that can be reserved.
It stores a unique identifier and the availability status. Two subtypes extend this entity: **Book** and **Room**.

Book

Specialization of Resource for library items.
It contains descriptive fields such as title, author, ISBN, publication year, and the number of copies available.

Room

Specialization of Resource for reservable spaces.
Includes attributes such as a room code, name, capacity, and physical location.

Reservation

This associative entity captures the act of reserving a resource by a client at a given time.
It stores the reservation identifier, date, status, time interval (start and end), and optional notes.

It connects clients and resources and may also be associated with an admin who oversees or approves it.

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2. Relationships

Client Reserves Resource

A client can make many reservations, and each resource can be reserved multiple times. Each reservation entry links one client to one resource and contains the details of that specific reservation.

Admin Controls Reservation

An admin may supervise or manage various reservations.
Each reservation can optionally be associated with one admin, indicating administrative control or approval.

Specialization Hierarchies

- **Resource → Book / Room**
  The system uses inheritance to represent different types of reservable items under a single parent entity.
- **User → Client / Admin**
  This hierarchy separates general user data from role-specific information, ensuring a clear structure for permissions and responsibilities.

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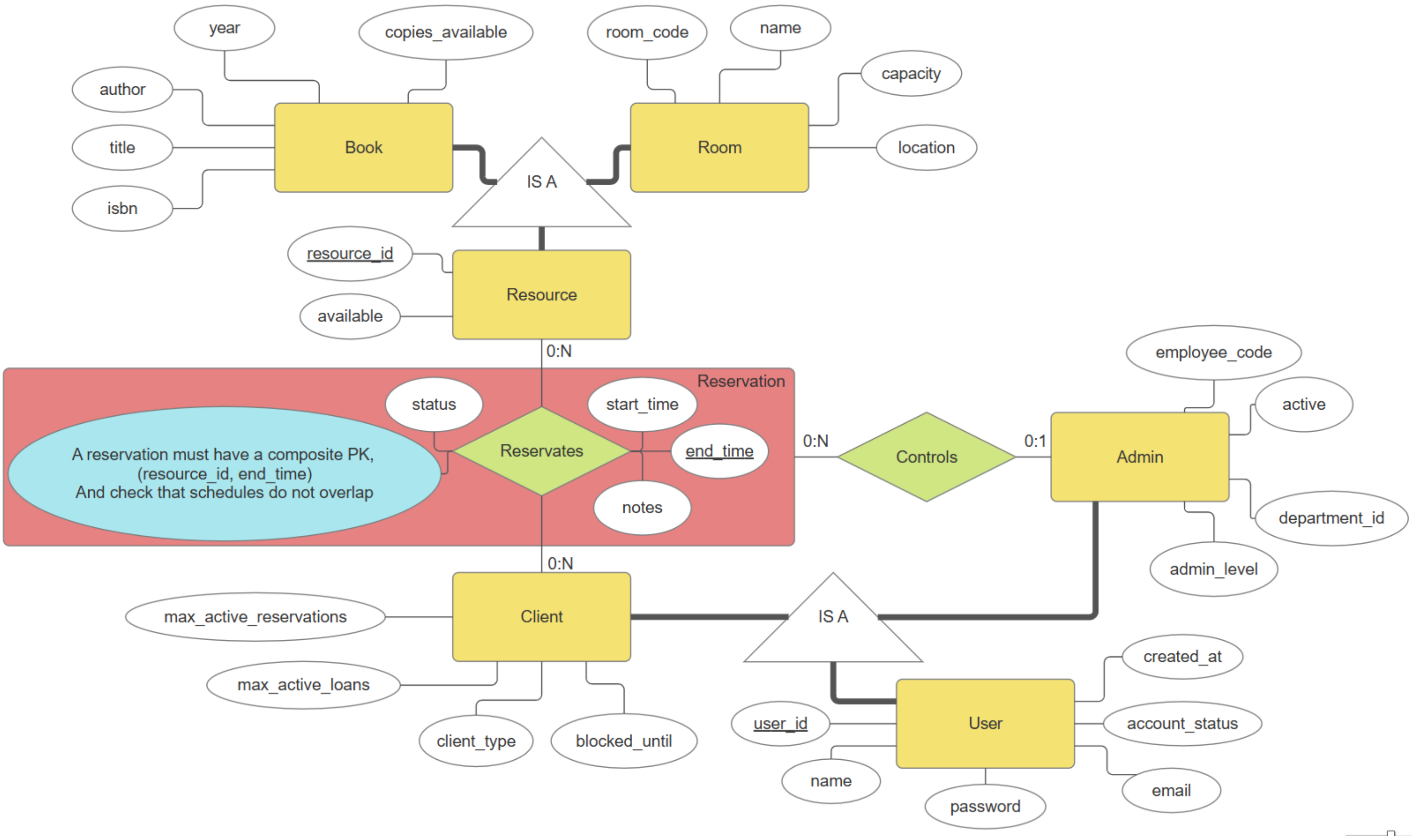
3. General Behavior of the System

The model supports:

- Creation, modification, and cancellation of reservations by clients.
- Administrative supervision of reservations, including approval workflows.
- Availability checking for books (copy count) and rooms (time slot).
- User lifecycle management for both clients and admins.

Overall, the model ensures that all reservation-related data is stored consistently while preserving a clear separation between resource types and user roles.

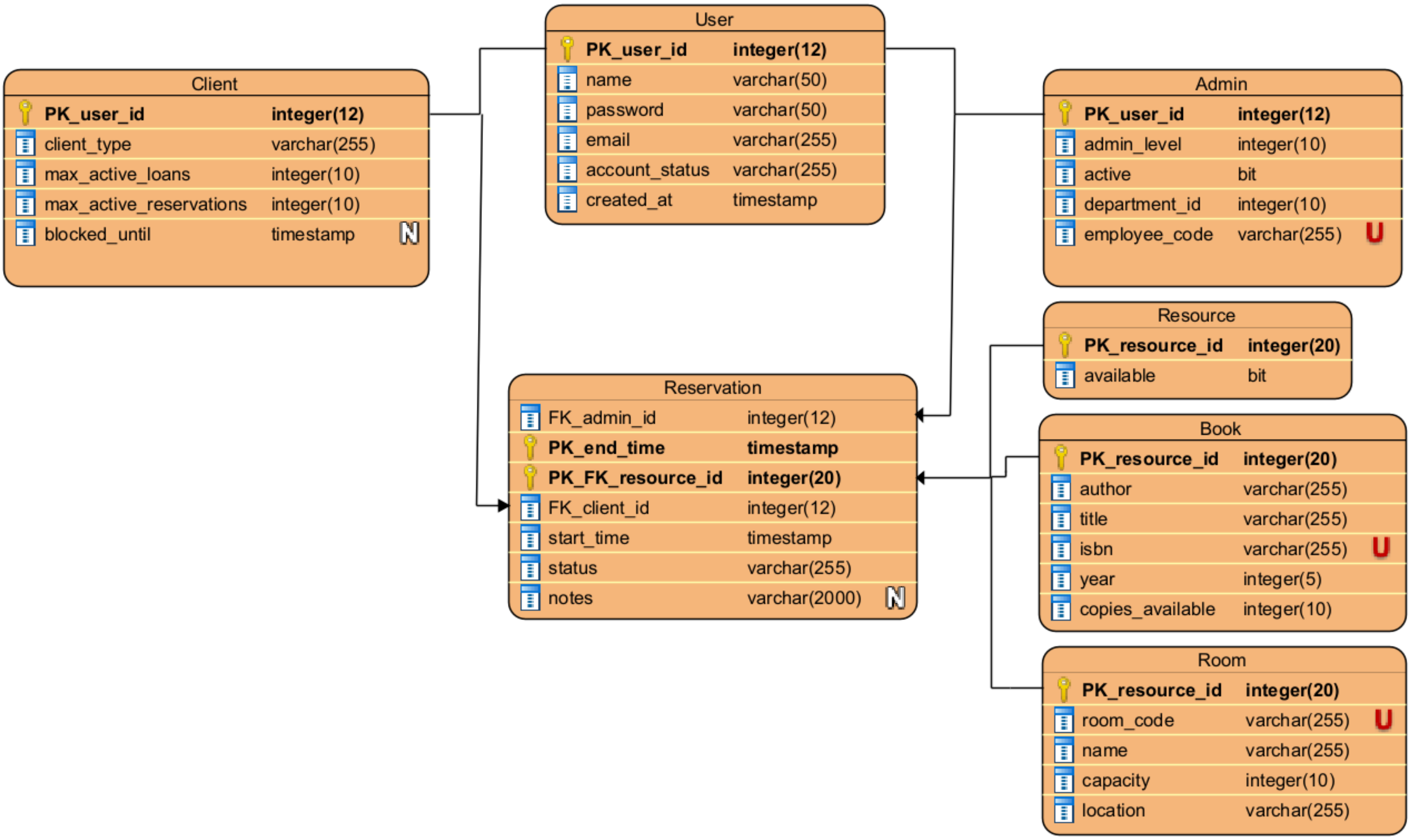
SE 2025 Winter - Project Report S3



3.3.2. Physical Model

The model presented as a diagram of the database that constitutes the implementation of the conceptual model in the selected database management system.

It is required to include an elaboration on the transformation of the conceptual model to the physical model. Description of transformation should include how elements like, e.g. inheritance and associations or other sophisticated modelling concepts have been handled.

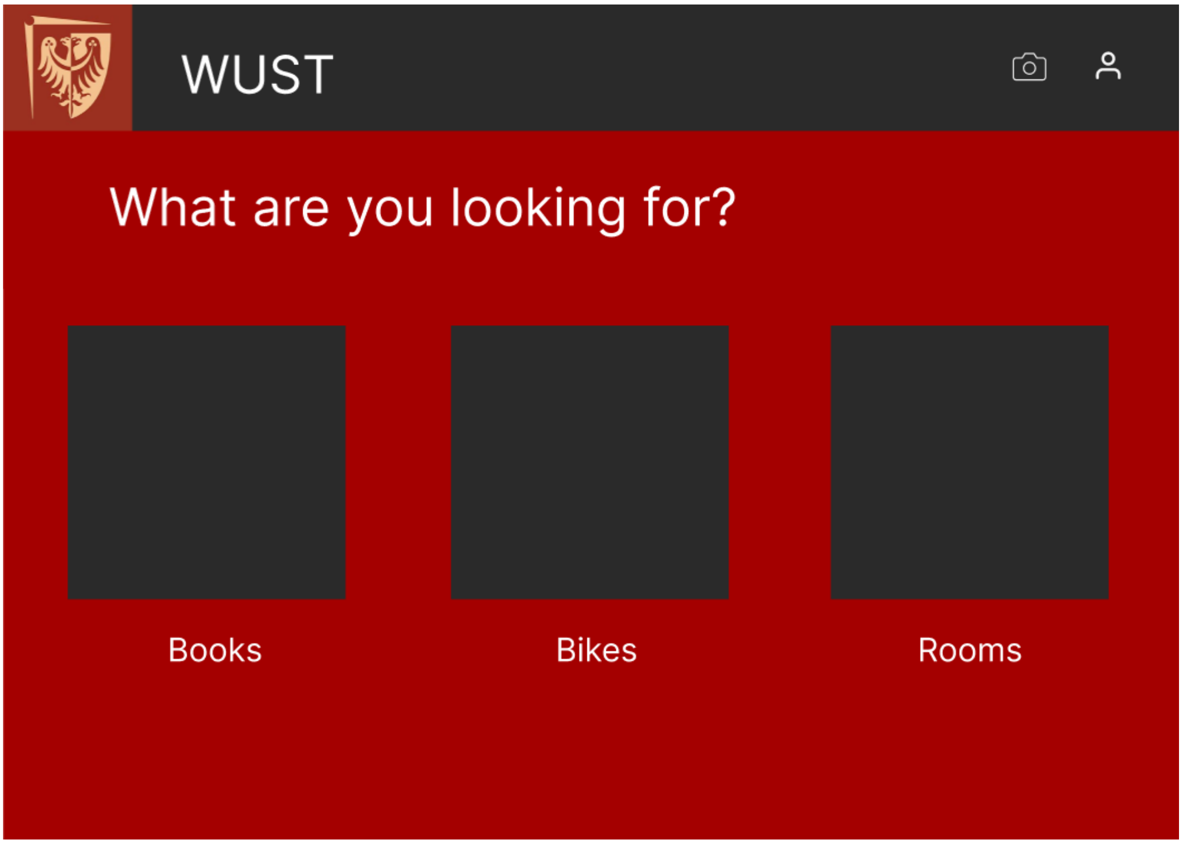


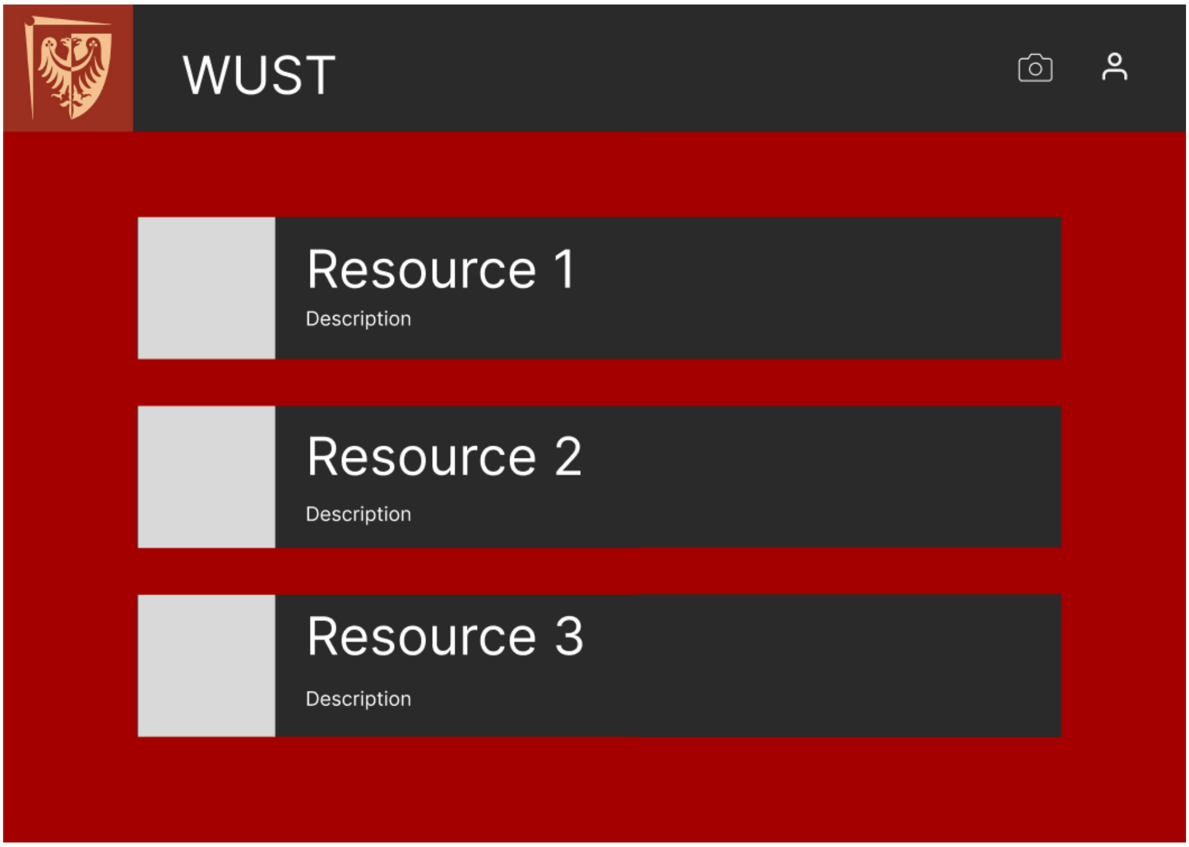
The User table is abstract as the thing that it does is saves writing items in diagram of the tables Client and Admin, the same happens with Resource, and Book and Room tables.

3.4. User Interface Design

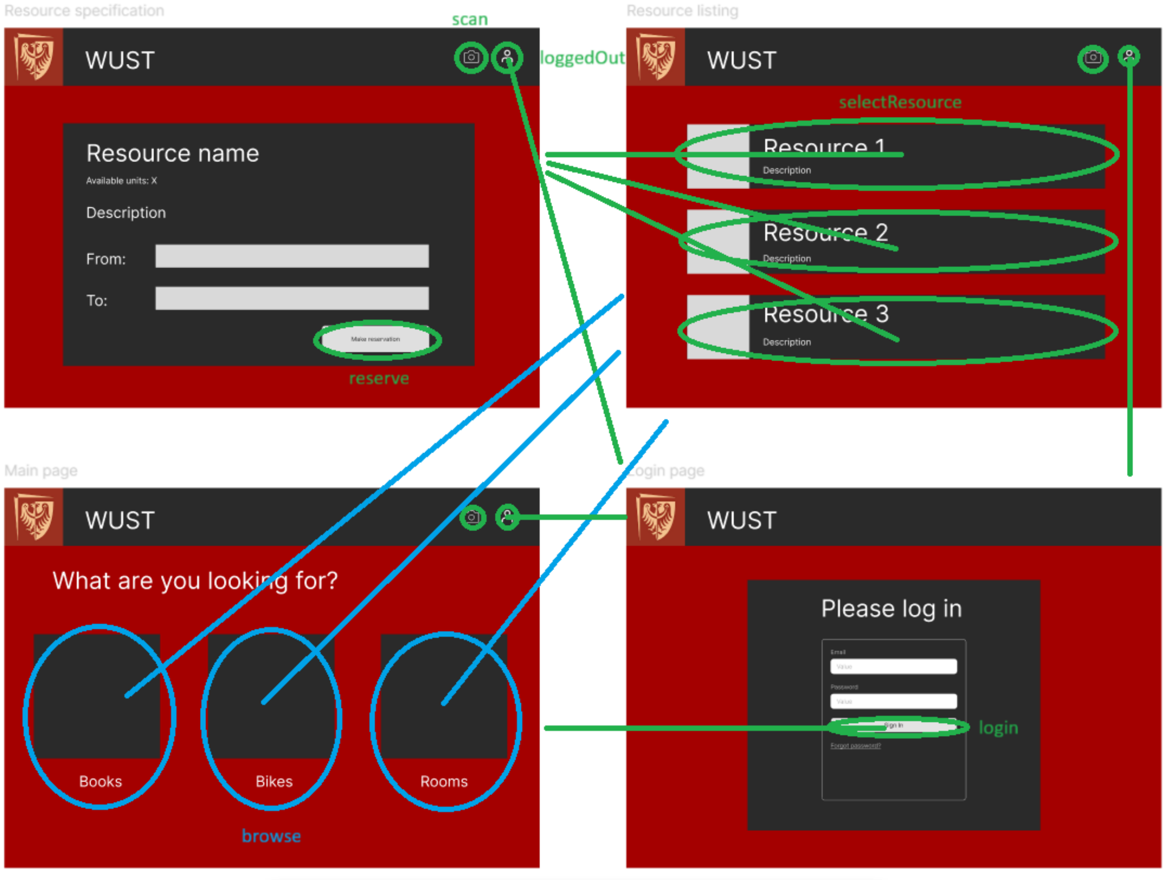
In this section, you should present the distribution of user interface components with events and their assignments to the behaviour specifications, which these elements initiate or take part.

These are the different pages we designed for this project, but we do not discard the fact that we may need to make new ones as the project advances and evolves.





Here is a description of the events that occur when the user interacts with said interface:



Note: All the events described in this picture are onclick events.